Panels of eight retained Suzuki markings

Human Compression project

Abstract

We verify the actual incidence panels of eight previously constructed markings of Sz(8). Together with the previously reviewed ninth marking, these give thin, flag-transitive, residually connected geometries for the nine indicated parameter pairs. This finite result does not assert that the markings exhaust the possible embeddings or prove the general thinness assertion in the source.

1 Matrices and the finite certificate

Use $\mathbb{F}_8 = \mathbb{F}_2[X]/(X^3 + X + 1)$, with integers $0, \dots, 7$ encoding the coefficients in the basis $1, X, X^2$. Thus the integer 2 denotes X, and addition is exclusive-or. In the classical Suzuki model put

$$w = \begin{pmatrix} 0 & 0 & 0 & 1 \\ 0 & 0 & 1 & 0 \\ 0 & 1 & 0 & 0 \\ 1 & 0 & 0 & 0 \end{pmatrix}, \qquad z(t) = \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ t & 0 & 1 & 0 \\ t^4 & t & 0 & 1 \end{pmatrix}, \quad h(t) = \operatorname{diag}(t^5, t, t^{-1}, t^{-5}).$$

We use the classical ambient matrix model and the classification of its maximal subgroups as explicit premises. The certificate checks the matrices, subgroups and cosets below directly, without enumerating the ambient group. The retained recipes, in their original order, are

(p, p')	$(2,5)$	$(2,7)$	$(2,13)$	$(5,7)$	$(5,13)$	$(7,7)$	$(7,13)$	$(13,13)$
x	1	4	3	1	1	4	2	3
y	–	–	–	2	3	–	3	–

For $p = 2$, take $(a, b, c) = (w, z(2), z(x))$ and $(G_0, G_1, G_2) = (\langle a, b\rangle, \langle b, c\rangle, \langle a, c\rangle)$. For $p = p'$ odd, take $(a, b, c) = (w, z(2)wz(2), z(x))$. In the remaining cases put $\lambda = (y^4x^{-4})^3$ and take $(a, b, c) = (w, h(\lambda)w, z(x))$. For both odd cases use $(G_0, G_1, G_2) = (\langle a, b\rangle, \langle a, c\rangle, \langle b, c\rangle)$.

The returned lists, checked independently by ordinary finite-field multiplication, have orders $14, 2p, 2p'$; all three pair intersections have order two and their triple intersection is trivial. The reader checks every product and inverse in each listed subgroup, and verifies that its two specified involutions generate the whole list. These data in particular identify the subgroups as the stated dihedral groups, with $D_4 = C_2 \times C_2$ when $p = 2$.

2 From the returned cosets to the geometry

For missing type i and other types j, k, list all distinct left cosets gG_i with $g \in G_j$, retaining precisely those that meet G_k. Every type-i object incident with G_j has a representative in G_j, so this procedure exhausts the actual panel above the edge (G_j, G_k). It uses at most thirteen candidates for any one panel. The certificate retains the entire coset and its intersections with G_j, G_k, and $G_j \cap G_k$, not only a claimed cardinality.

For each of the eight recipes and each of the three missing types, the independent reader obtains exactly two cosets, both meeting $G_j \cap G_k$. There are thus twenty-four panels of size two, and every chamber containing a base edge has a common representative.

Proposition 1. *Each of the eight coset geometries is thin, flag transitive and residually connected. It satisfies RWPRI and local* 2*-transitivity under the literal algebraic-residue convention of the fixed source.*

Proof. Every incident pair has a common representative, so it translates to a base edge. The calculation therefore proves that every panel has size two. It also proves that every chamber translates to the base chamber: after translating one of its edges to (G_j, G_k), the remaining coset meets $G_j \cap G_k$. This proves flag transitivity; every flag extends to a chamber by the common-representative argument in rank three.

Each G_i is generated by its two distinct order-two intersections with the other parabolics. Thus its rank-two residue is connected: successive multiplication by the two intersection subgroups gives an incidence walk to every coset. The whole incidence graph is connected as well. Indeed $G_0 = D_{14}$ is maximal in $\mathrm{Sz}(8)$, and an additional parabolic has intersection of order two with G_0, so the three parabolics generate the ambient group. The coset-walk argument applies again. Rank-one residues impose no further connectedness requirement.

For the empty flag, G_0 supplies the required maximal subgroup. In each parabolic, either of its order-two intersection subgroups is maximal, since its index is one of $2, 5, 7, 13$. For a base edge the trivial triple intersection is maximal in its order-two stabilizer. These are all the RWPRI conditions. Finally the algebraic rank-one residues are the regular two-point actions of the pair intersections, and hence are 2-transitive. □

Scope and attribution. The constructions and the earlier $(5, 5)$ example are retained phase-five work; the latter already has a distinct accepted panel certificate. The new contribution here is the complete literal incidence check and its application for the other eight saved markings. The statement concerns these markings, not all embeddings with the same orders. The source is D. Leemans, *The rank 3 geometries of the simple Suzuki groups* $\mathrm{Sz}(q)$, Note di Matematica 19(1) (1999), 43–63, especially its definitions on printed pp. 45–46 and Theorem 6.1 on p. 61. The source's algebraic residue definition and actual thinness conclusion remain distinct notions until proved to agree in an application, as above. There is no new source-completion, refutation, conjecture, Lean or compression credit from these positive checks.